THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A constant force of 50 N displaces a body by 3 m in the direction of the force. The work done is: 1

(a) 300 J **(b)** 16.67 J **(c)** 53 J **(d)** 150 J

2. The kinetic energy of a body of mass 10 kg moving at 6 m s^{-1} is: 1

(a) 60 J **(b)** 90 J **(c)** 180 J **(d)** 360 J

3. The potential energy of a 5 kg body at a height of 10 m (g = 10 m s^{-2}) is: 1

(a) 1000 J **(b)** 50 J **(c)** 250 J **(d)** 500 J

4. A machine does 1500 J of work in 30 s. Its power is: 1

(a) 50 W **(b)** 100 W **(c)** 0.02 W **(d)** 45000 W

5. A machine lifts a load of 500 N by applying an effort of 125 N. Its mechanical advantage is: 1

(a) 5 **(b)** 4 **(c)** 0.25 **(d)** 8

6. On a lever the load is 80 N at 10 cm from the fulcrum. To balance it with the effort arm 40 cm, the effort needed is: 1

(a) 40 N **(b)** 90 N **(c)** 80 N **(d)** 20 N

7. Work done by a goalkeeper in stopping a ball is: 1

(a) positive **(b)** zero **(c)** negative **(d)** infinite

8. A fixed pulley is used to: 1

(a) increase the work done **(b)** reduce the effort
(c) change the direction of the effort **(d)** reduce the load

9. The work-energy theorem states that the work done on a body equals the change in its: 1

(a) momentum only **(b)** force **(c)** mass **(d)** energy

10. The speed of a child reaching the bottom of a smooth slide of height h depends on: 1

(a) the mass of the child **(b)** h only (and g)
(c) the time taken **(d)** the shape of the slide

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. When is the work done negative? Give an example. 2

12. State the work-energy theorem. 2

13. Write the expressions for kinetic and gravitational potential energies. 2

14. Calculate the kinetic energy of a 0.2 kg ball moving at 10 m s^{-1}. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Derive the expression for gravitational potential energy. 3

16. A fielder throws a 200 g ball 10 m high. Find its PE at the highest point and its speed on returning to the ground (g = 10). 3

17. Show that the mechanical energy is conserved for a freely falling body at the initial and final points. 3

OR

Explain the energy changes in a swinging pendulum. 3

18. A child slides down a smooth slide of height 2.5 m. Find the speed at the bottom (g = 10). 3

OR

Name the energy transformation in: 3
(i) a glowing bulb
(ii) a ringing bell
(iii) photosynthesis.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A ball of mass 0.5 kg is dropped from 20 m (g = 10). 5
(i) Find its initial PE.
(ii) Find its speed on reaching the ground.
(iii) Find its KE at 10 m above the ground.
(iv) State the principle used.

OR

(i) Describe the pendulum activity for conservation of energy. 5
(ii) Why does a real pendulum stop?
(iii) Name the energy at the extreme and mean positions.

20. (i) Define power and write its SI unit. 5
(ii) Differentiate work and power.
(iii) A motor lifts 120 kg through 15 m in 20 s. Find its power (g = 10).

OR

Case study: A student of mass 40 kg is lifted by a lift to the top of a 72.5 m building, then later climbs the stairs. 5
(i) Find the gain in PE in the lift (g = 10).
(ii) Find the gain by stairs.
(iii) What do you conclude?
(iv) If the stair climb takes 100 s, find the power.

END OF QUESTION PAPER — **Total: 40 Marks**